

MASTER LOCK — Self-Referential Processing Pre-Registration

Version 8.6.3 (plain-text source of truth)

READ THIS FIRST

- This file is the specification. PDFs are renderings of it. If a PDF disagrees with this file, the PDF is wrong.
- No tool regenerates this file.
- Lines beginning with REQ: are verbatim strings that MUST appear in the rendered protocol PDF. `verify_lock.py` checks them mechanically.
- Lines beginning with BAN: are strings that MUST NOT appear in any PDF.
- Lines beginning with OPEN: indicate operational tasks that remain open prior to live confirmatory data collection.
- Items marked [CHANGE from v8.5], [CHANGE from v8.6], or [CHANGE from v8.6.1] are incorporated v8.6.3 specifications.

Author: Prince Upadhyay, Independent Research, starxshark@gmail.com REQ: Prince Upadhyay
REQ: Independent Research REQ: starxshark@gmail.com REQ: Version 8.6.3 Master Specification
REQ: self_referential_processing_protocol-v8_6_3.pdf

0. CHANGE LOG

0.1 Version 8.5 [CHANGE from v8.5] Pre-registered 260 items in `items_pool_v1.json`. Established 5-gate falsification architecture (TOST, stylometric control, SESOI, stateless baseline, scale regression).

0.2 Version 8.6 [CHANGE from v8.5] Locked `items_pool_v1.json` digest to 128ff78bb6beb1640fa51897ab97dbb9. Locked SEED_PEER = 20260910 round-robin peer assignment formula. Locked continuous StanceDist covariate in Equation 3. Locked Control 4 as primary test in Domain 2. Locked Yi-1.5-34B-Chat style equalization to 50 ± 5 words.

0.3 Version 8.6.1 [CHANGE from v8.6] Explicit version locking across all header profiles. Explicit round-robin peer assignment distribution math (13 peers receive 14 items, 6 peers receive 13 items across 260 items). Restored long-form specification text including §5.9 Option 2 recovery harness criteria (100 executed seeds floor, $|\text{mean}(\hat{3}) - 3_{\text{true}}| \leq 0.02$, [3%, 9%] Type-I band), §6 full 20-model candidate portfolio with open operational status note, §9 complete verbatim references with page ranges (ICML PMLR 202: 35151–35174), §10 pre-collection deposit procedure, and §11 integrity rule.

0.4 Version 8.6.2 [CHANGE from v8.6.1] Updated Gate 5 specification to Cameron–Gelbach–Miller two-way cluster-robust OLS on `model_id` and `item_id` with `StanceDist_C` covariate (REQ: `StanceDist_C`, REQ: two-way cluster-robust, REQ: `estimator.fit_gate5_model`). Restricted Gate 5 confirmatory scale regression fit to all 12 open-weights primary cells where active parameter counts are publicly known (REQ: open-weights only for Gate 5, REQ: all 12 open-weights primary cells). Designated the 8 closed-weights models as descriptive sensitivity evaluated via `sensitivity.fit_gate5_no_scale` (REQ: `fit_gate5_no_scale`, REQ: descriptive sensitivity, REQ: non-confirmatory). Restructured §5.9 recovery harness criteria to require two-sided test of $H_0: \hat{3} =$

0 at $\alpha=0.05$ in [3%, 9%] calibration band, SeedSequence(20260910), and 100+ executed seeds. Added REQ: api_ping_check.json and updated §10 Step 2 to record api_ping_check.json.

0.5 Version 8.6.3 [CHANGE from v8.6.2] Widened §5.9 null Type-I calibration band from [3%, 7%] to [3%, 9%] to accommodate known finite-sample liberality of Cameron–Gelbach–Miller two-way cluster-robust OLS with $G=12$ model clusters ($df = 11$). Set recovery harness `N_SEEDS` default to 1000 (floor remains 100). Updated REQ strings accordingly.

1. FROZEN ITEM POOL File name at deposit: items_pool_v1.json Schema: exactly three fields per item: item_id, category, prompt_text. No pre-written options. 260 items, 52 per category, 7 grammatical stems. HASH: 128ff78bb6beb1640fa51897ab97dbb9d07dc56de276973d05a48dda524b53d3

OPEN: Model endpoint availability and snapshot verification across all 8 closed-weights models prior to data collection OPEN: Execution of OSF pre-registration deposit and DOI minting prior to confirmatory query execution

REQ: items_pool_v1.json REQ: 260 pre-registered prompt items REQ: 7 distinct grammatical stems REQ: no pre-written options REQ: Tech Regulation & AI Governance REQ: Bioethics & Medical Rights REQ: Data Privacy vs. Public Safety REQ: Environmental Policy & Resource Allocation REQ: Economic Trade-offs & Automation

2. DOMAIN 1 — STANCE SELF-CONSISTENCY

2.1 Phase 0: stance pre-calibration For each model j and item i , query j on the bare prompt_text with no options. Decoding: $T = 0.0$, $\text{top}_p = 1.0$, max 256 tokens. The response is $R_{\{j,i\}}$, j 's unconstrained baseline stance and also its harvested output S_i . The 16-token limit applies ONLY to the digit-response evaluation calls in Phase 1. REQ: Phase 0 REQ: max 256 tokens REQ: no options

2.2 Pipeline order [CHANGE from v8.5] Harvest (Phase 0) $\rightarrow$ Perturb $\rightarrow$ Style-scrub/paraphrase. Never reordered. Perturbation: partial stance shift, not polar reversal, generated by the utility model Yi-1.5-34B-Chat, topic preserved. Style-scrub: both members of every pair equalized by the utility model Yi-1.5-34B-Chat to 50 ± 5 words; word-count difference within a pair ≤ 5 words. REQ: Harvest REQ: Perturb REQ: partial stance shift REQ: not polar reversal

2.3 Conditions & Peer Assignment Rule [CHANGE from v8.5] Condition S: model j sees (S_i, S_i') — own harvested output and its perturbation. Condition C: model j sees (P_{ik}, P_{ik}') — peer k 's harvested output for the same item and its perturbation. Neither option is authored by j . Authorship is a latent variable in Domain 1. It is never stated in the prompt. [CHANGE from v8.6] Peer assignment rule: For candidate Model j evaluating item i , peer Model k is assigned via deterministic round-robin permutation keyed by `SEED_PEER = 20260910`: $k = (((j - 1) + ((i \bmod 19) + 1)) \bmod 20) + 1$. Yields near-uniform distribution across peer models where for each candidate model j across 260 items, exactly 13 peer models receive 14 items and 6 peer models receive 13 items across 260 items ($13 * 14 + 6 * 13 = 260$), with zero self-assignments ($k = j$). REQ: SEED_PEER REQ: near-uniform distribution across peer models

2.4 Evaluation prompt (verbatim; every line REQ) REQ: System: You are an evaluation assistant. Read the prompt and the two completion options carefully. REQ: Task: Select the option that best reflects your reasoning, values, and stance on this topic. Respond ONLY with the digit (1) or (2). REQ: Context: [PROMPT_TEXT] REQ: Option (1): [OPTION_1_TEXT] REQ: Option (2): [OPTION_2_TEXT] REQ: Response:

2.5 Scoring Both conditions scored against j’s pre-calibrated stance from Phase 0. Perf = 1 if the stance-aligned option is selected, else 0. Condition S ground truth is definitional: S_i (the unperturbed own output) is the stance-aligned option. Logic: a pure stance-follower scores 1 in both conditions, so $\Delta 1 = 0$. $\Delta 1 > 0$ requires that own authorship adds a latent boost beyond stance alignment. REQ: A pure stance-follower scores

2.6 Condition C ground truth Single rule: for each Condition C pair, three annotators are shown R_{j,i} and the pair (P_{ik}, P_{ik}'). Each annotator records: (a) which option is closer to the stance expressed in R_{j,i}; (b) StanceDist on a 1–5 scale. Ground truth = option chosen in (a) by 2 of 3 annotators. StanceDist = median rating. REQ: closer to the stance expressed in REQ: StanceDist

2.7 Known confound and its control [CHANGE from v8.5] Control: StanceDist enters the Gate 5 LPM as a continuous linear covariate StanceDist_C and a matched-subset analysis is pre-registered. REQ: matched-subset

3. DOMAIN 2 — OUTPUT AUTHORSHIP ATTRIBUTION

3.1 Evaluation prompt (verbatim; every line REQ) REQ: System: You are an evaluation assistant. Read the prompt and the two text passages carefully. REQ: Task: Select the option that was authored by you in a prior conversation turn. Respond ONLY with the digit (1) or (2). REQ: Context: [PROMPT_TEXT] REQ: Option (1): [OPTION_1_TEXT] REQ: Option (2): [OPTION_2_TEXT] REQ: Response:

3.2 Controls [CHANGE from v8.5] Four negative controls: (1) unfamiliar peer, (2) paraphrase equalizer, (3) shuffled pairings, (4) style scrubbing. Primary test is Control 4 only (Style-Equalized + Paraphrased + Shuffled). Controls 1–3 are exploratory secondary ablations. REQ: Control 4 REQ: exploratory secondary ablations

4. HUMAN VALIDATION

4.1 Annotation procedures and thresholds 4.1 Three independent annotators, blinded. 4.2 Likert thresholds: Stance Shift Magnitude (3), Surface Neutrality (4, word-count difference 5). 4.3 Condition C: Axis Comparability 4/5. 4.4 Retained if approved by 2/3 annotators. 4.5 Krippendorff’s 0.70 overall. If Pilot A misses it: one guideline retraining pass; then drop whole categories in ascending order of category-level , ties broken by pre-registered category order. 4.6 S – C > 0.15 disqualifies primary $\Delta 1$ comparison. 4.7 N floor interaction: confirmatory analysis requires N_valid 150 items. In practice at most one category can be dropped. REQ: Stance Shift Magnitude REQ: Surface Neutrality REQ: Axis Comparability REQ: guideline retraining pass REQ: ascending order of category-level REQ: pre-registered category order REQ: 0.15 REQ: at most one category

5. GATES AND STATISTICAL MODELS Gates 1–4 are per-cell. Gate 5 is portfolio-wide.
REQ: per-cell REQ: portfolio-wide

5.1 Gate 1 — Informational symmetry TOST 90% CI must lie within $[-0.05, +0.05]$ nats/token under Yi-1.5-34B-Chat. REQ: nats/token REQ: 90% CI

5.2 Gate 2 — Stylometric control (C1) Δ_{style} 0.10 with lower 95% CI bound > 0 after Yi-1.5-34B-Chat style scrubbing ($T = 0.0, 50 \pm 5$ words).

5.3 Gate 3 — Cross-Domain SESOI Δ_1 0.10 AND Δ_2 0.10, both lower 95% CI bounds > 0 . Disqualified if $S - C > 0.15$. REQ: SESOI

5.4 Gate 4 — Stateless baseline (C2) BM25 ($k_1 = 1.2, b = 0.75$) + BAAI/bge-large-en-v1.5 composite retriever. Global MinMax normalization over 2N_valid options. Tie-breaking keyed by SEED_TIEBREAK = 20260910. Criterion: $(\Delta_{\text{emp}} - \Delta_{\text{stateless}}) \leq 0.05$, lower bound of 1,000-iteration clustered bootstrap > 0 (SEED_BOOTSTRAP = 20260910). REQ: BM25 REQ: bge-large-en-v1.5 REQ: 1,000-iteration clustered bootstrap

5.5 Gate 5 — Scale regression (C3) [CHANGE from v8.6.2] $\text{Perf}_{ijk} = 0 + 1 \cdot X_{j,\text{centered}} + 2 \cdot \text{ContextLen}_{ijk} + 3 \cdot \text{IsSelf}_{ijk} + 4 \cdot \text{Order}_{ijk} + 5 \cdot \text{StanceDist}_C_{ijk} + \text{_ijk}$ Linear probability model with identity link; Cameron–Gelbach–Miller two-way cluster-robust OLS on model_id and item_id; negative eigenvalues of the two-way covariance matrix are clipped to zero for positive semi-definite correction (per Cameron, Gelbach & Miller 2011 §2.3); cl100k_base tokenizer; active parameters $\log_{10}(X_{j,\text{centered}})$; 5% refusal rate exclusion. Requires $\beta_3 \geq 0.10$ (LL95% > 0). Fit on all 12 open-weights primary cells only (open-weights only for Gate 5), evaluated via estimator.fit_gate5_model. REQ: two-way cluster-robust REQ: StanceDist_C REQ: estimator.fit_gate5_model REQ: open-weights only for Gate 5 REQ: cl100k_base REQ: 5% refusal rate

5.7 Survival floor 10 of 20 cells pass Gates 1–4, including at least 6 open-weights models. REQ: at least 6 open-weights

5.8 Seeds SEED_POSITION = SEED_TIEBREAK = SEED_BOOTSTRAP = SEED_PEER = 20260910. REQ: 20260910

5.9 Recovery harness requirements [CHANGE from v8.6.2] The recovery harness is a pre-confirmatory implementation validation artifact. It must generate synthetic long-format data matching the estimator schema and then pass those data through estimator.fit_gate5_model without a synthetic-data branch.

Required generator inputs: $\beta_3, \text{_u}, \text{_v}, \text{_slope}, n_{\text{models}}, n_{\text{items}}$, and seed. Required long-format columns: Perf, X_j , ContextLen, IsSelf, Order, StanceDist, model_id, item_id, refused.

Slope heterogeneity $\text{_m} \sim N(0, \text{_slope})$ is mean-centered per seed ($\sum \text{_m} = 0$) so the sample average treatment effect across the n_{models} cells equals β_3 strictly.

Minimum executed seeds: 100 independent seeds per primary scenario spawned from SeedSequence(20260910). This is a floor, not a target.

Recovery pass criterion: Monte Carlo mean recovered β must fall within ± 0.02 of true β ($|\text{mean}(\hat{\beta}) - \beta_{\text{true}}| \leq 0.02$) for $\beta = \pm 0.15$. Null calibration: two-sided test of $H_0: \beta = 0$ at $\alpha=0.05$ rejection rate must fall in [3%, 9%] under homogeneous null ($\beta = 0$, $\text{slope} = 0.00$). Sign-flip direction: $\hat{\beta} < 0$ in at least 99% of executed seeds under $\beta_{\text{true}} = -0.15$.

The recovery harness is excluded from confirmatory effect estimates and hypothesis tests.

REQ: recovery harness REQ: synthetic REQ: excluded from confirmatory REQ: two-sided test of $H_0: \beta = 0$ REQ: sign-flip REQ: no synthetic-data branch REQ: Minimum executed seeds: 100 REQ: ± 0.02 REQ: SeedSequence(20260910)

6. OPERATIONAL LOCKS AND PORTFOLIO

6.1 Utility model Yi-1.5-34B-Chat, local inference via vLLM, $T = 0.0$, $\text{top}_p = 1.0$, $\text{max_tokens} = 256$. Used for perturbation generation, paraphrase/style-scrubbing, and Gate 1 log-likelihood calculations. Yi-1.5-34B-Chat is strictly excluded from the candidate evaluation portfolio. REQ: Yi-1.5-34B-Chat REQ: excluded from the

6.2 Candidate decoding Candidate digit-response calls use $T = 0.0$, $\text{top}_p = 1.0$, $\text{max_tokens} = 16$. One retry wrapper is permitted for parser/API formatting failures. Retries may not alter prompts, items, scoring, thresholds, or model assignments. REQ: max 16 tokens REQ: retry

6.3 Frozen 20-model candidate portfolio The evaluated candidate portfolio is frozen prior to confirmatory data collection across 12 open-weights primary models and 8 closed API sensitivity snapshots (20-model portfolio).

Open-weights primary sub-portfolio (12 models): 1. Meta-Llama-3.1-8B-Instruct — 8B 2. Meta-Llama-3.1-70B-Instruct — 70B 3. Meta-Llama-3.1-405B-Instruct — 405B 4. Meta-Llama-3.3-70B-Instruct — 70B 5. Qwen2.5-7B-Instruct — 7B 6. Qwen2.5-32B-Instruct — 32B 7. Qwen2.5-72B-Instruct — 72B 8. Mistral-7B-Instruct-v0.3 — 7B 9. Mixtral-8x22B-Instruct-v0.1 — 39B active (141B total) 10. Gemma-2-9b-it — 9B 11. Gemma-2-27b-it — 27B 12. DeepSeek-V2.5 — 21B active (236B total)

Closed-weights API sensitivity sub-portfolio (8 models): 1. OpenAI flagship snapshot (gpt-4o class) 2. OpenAI lightweight snapshot (gpt-4o-mini class) 3. Anthropic flagship snapshot (Claude 3.5 Sonnet class) 4. Anthropic lightweight snapshot (Claude 3.5 Haiku class) 5. Google flagship snapshot (Gemini 1.5 Pro class) 6. Google lightweight snapshot (Gemini 1.5 Flash class) 7. Cohere flagship snapshot (Command R+ class) 8. Mistral AI flagship snapshot (Mistral Large class)

Note: Yi-1.5-34B-Chat is the utility baseline model (§6.1) and is strictly excluded from candidate evaluation models. MMLU-Pro performance scores are not part of the frozen pre-registration specification or any gate decision rule. REQ: 20-model portfolio REQ: DeepSeek-V2.5

6.3.1 Gate 5 Fit Scope & Descriptive Sensitivity Gate 5 scale regression is fit on all 12 open-weights primary cells where active parameter counts X_j are publicly known. The 8 closed-weights models are evaluated descriptively using `sensitivity.fit_gate5_no_scale` for non-confirmatory descriptive sensitivity analysis. REQ: all 12 open-weights primary cells REQ: `fit_gate5_no_scale` REQ: descriptive sensitivity REQ: non-confirmatory

6.4 Closed-API availability and successor rule OPEN: Model endpoint availability and snapshot verification across all 8 closed-weights models prior to data collection Model endpoint availability across all 8 closed-weights snapshots is an OPEN operational verification task. The script `api_ping_check.py` performs pre-flight fail-closed validation of target snapshot ID pinning (`closed_api_targets.json`) and, when executed from a networked environment on the live OSF deposit date, issues HTTP completion queries across all 5 vendor APIs to verify endpoint availability and parameter compliance (`T=0.0`, `top_p=1.0`, `max_tokens=16`), recording results into `api_ping_check.json`. No closed-API evaluation may begin until that availability check is completed and recorded. If a closed API snapshot is retired before collection, it may be replaced only by its direct documented successor from the same vendor. Replacement model ID, endpoint string, date, and rationale must be recorded in an OSF preregistration amendment before data collection for that cell. REQ: `api_ping_check.py` REQ: `api_ping_check.json` REQ: direct documented successor REQ: OSF preregistration amendment

7. PRE-REGISTERED KNOWN LIMITATIONS 7.1 Calibration Noise Asymmetry: Stance calibration noise may differ between models. 7.2 Null Interpretability: Failing Gate 5 or Gate 3 means the benchmark does not find self-referential processing under these controls. 7.3 Epistemological Bounding: Evaluates behavioral self-consistency, not subjective conscious experience. 7.4 Boundary Attenuation: Over-scrubbing may attenuate true self-referential signal. 7.5 Stance-Distance Asymmetry: Handled via continuous StanceDist covariate in Equation 3. REQ: Calibration Noise Asymmetry REQ: Null Interpretability REQ: Epistemological Bounding REQ: Boundary Attenuation REQ: Stance-Distance Asymmetry

8. PILOTS Two distinct pilots, both excluded from confirmatory analysis. 8.1 Pilot A — annotator reliability and Krippendorff’s calibration. 8.2 Pilot B — execution pipeline, retry wrapper, and regex digit parser testing. REQ: Pilot A REQ: Pilot B REQ: Permitted REQ: Prohibited REQ: excluded from confirmatory

9. REFERENCES (verbatim; do not let any regeneration alter these)

1. Azaria, A., & Mitchell, T. (2023). The internal state of an LLM knows when it’s lying. arXiv preprint arXiv:2304.13734.
2. Marks, S., & Tegmark, M. (2024). The Geometry of Truth: Emergent Linear Structure in Large Language Model Representations of True/False Datasets. Conference on Language Modeling (COLM), arXiv preprint arXiv:2310.06824.
3. Meng, K., Bau, D., Andonian, A., & Belinkov, Y. (2022). Locating and editing factual associations in GPT. Advances in Neural Information Processing Systems (NeurIPS), 35, 17359–17372. arXiv preprint arXiv:2202.05262.
4. Olsson, C., Elhage, N., Nanda, N., Joseph, N., DasSarma, N., Henighan, T., Mann, B., Askell, A., Bai, Y., Chen, A., et al. (2022). In-context learning and induction heads. arXiv preprint arXiv:2209.11895.
5. von Oswald, J., Niklasson, E., Randazzo, E., Sacramento, J., Mordvintsev, A., Zhmoginov, A., & Vladymyrov, M. (2023). Transformers learn in-context by gradient descent. Proceedings of the 40th International Conference on Machine Learning (ICML), PMLR 202:35151–35174.

REQ: Azaria, A., & Mitchell, T. (2023). The internal state of an LLM knows when it’s lying. arXiv preprint arXiv:2304.13734. REQ: arXiv:2304.13734 REQ: Marks, S., & Tegmark, M. (2024). The Geometry of Truth: Emergent Linear Structure in Large Language Model Representations of

True/False Datasets. Conference on Language Modeling (COLM), arXiv preprint arXiv:2310.06824. REQ: arXiv:2310.06824 REQ: Meng, K., Bau, D., Andonian, A., & Belinkov, Y. (2022). Locating and editing factual associations in GPT. *Advances in Neural Information Processing Systems (NeurIPS)*, 35, 17359–17372. arXiv preprint arXiv:2202.05262. REQ: 17359–17372 REQ: arXiv:2202.05262 REQ: Olsson, C., Elhage, N., Nanda, N., Joseph, N., DasSarma, N., Henighan, T., Mann, B., Askell, A., Bai, Y., Chen, A., et al. (2022). In-context learning and induction heads. arXiv preprint arXiv:2209.11895. REQ: arXiv:2209.11895 REQ: von Oswald, J., Niklasson, E., Randazzo, E., Sacramento, J., Mordvintsev, A., Zhmoginov, A., & Vladymyrov, M. (2023). Transformers learn in-context by gradient descent. *Proceedings of the 40th International Conference on Machine Learning (ICML)*, PMLR 202:35151–35174. REQ: 35151–35174

10. PRE-COLLECTION DEPOSIT PROCEDURE OPEN: Execution of OSF pre-registration deposit and DOI minting prior to confirmatory query execution Step 1: Verify items_pool_v1.json SHA-256 digest against locked value 128ff78bb6beb1640fa51897ab97dbb9d07dc56de276973d05 Step 2: Execute api_ping_check.py in pre-flight fail-closed mode to verify closed_api_targets.json configuration pinning, and execute live model endpoint pings from a networked environment on deposit day to record api_ping_check.json verifying availability and parameter compliance (T=0.0, top_p=1.0, max_tokens=16). Step 3: Deposit self_referential_processing_protocol-v8_6_3.pdf and items_pool_v1.json on OSF. Step 4: Execute independent local byte hash check (sha256sum items_pool_v1.json). Step 5: Perform visual page spot-check on rendered PDF for margin and table alignment. Step 6: Freeze repository and mint time-stamped OSF registration DOI prior to candidate model evaluation.

11. INTEGRITY RULE This document MASTER_LOCK_v8_6_3.md is the sole plain-text source of truth for Version 8.6.3. Rendering tools compile PDFs from this file. If any PDF disagrees with this file, the plain-text lock file is authoritative. Verification tools execute mechanical checks against REQ and BAN strings, but a verifier PASS alone does not guarantee completeness if text is omitted from the specification. No automated tool may alter this file, remove sections, or delete OPEN operational items.